About the Project: AI Embryo Quality Assessment & Grading

Clinical Background, System Overview Block Diagram & Architecture Guide for Web Development

1. Clinical Motivation & Project Goal

In In Vitro Fertilization (IVF), selecting the highest-competence blastocyst for single embryo transfer (eSET) is the critical determinant of implantation success and healthy live birth. Currently, embryologists manually evaluate blastocysts under inverted optical microscopes using subjective morphological criteria (Gardner & Schoolcraft 1999; ALPHA/ESHRE Consensus 2011). However, manual grading suffers from documented inter-observer disagreement between 30% and 40%, introducing substantial clinical variability.

Project Objective: Engineer an objective, automated Deep Learning System that categorizes human blastocysts into three clinical tiers (**Grade-1 Good**, **Grade-2 Fair**, **Grade-3 Poor**) and delivers immediate viability predictions (Transferable vs Non-Viable) with zero patient data leakage.

2. System Overview & Architecture (Figure 1: Block Diagram)

The end-to-end clinical workflow consists of multi-focal microscopic image acquisition, adaptive image preprocessing, a 4-backbone heterogeneous deep ensemble with 5-view test-time augmentation, and dual clinical decision heads (3-Class Grade + Binary Viability):

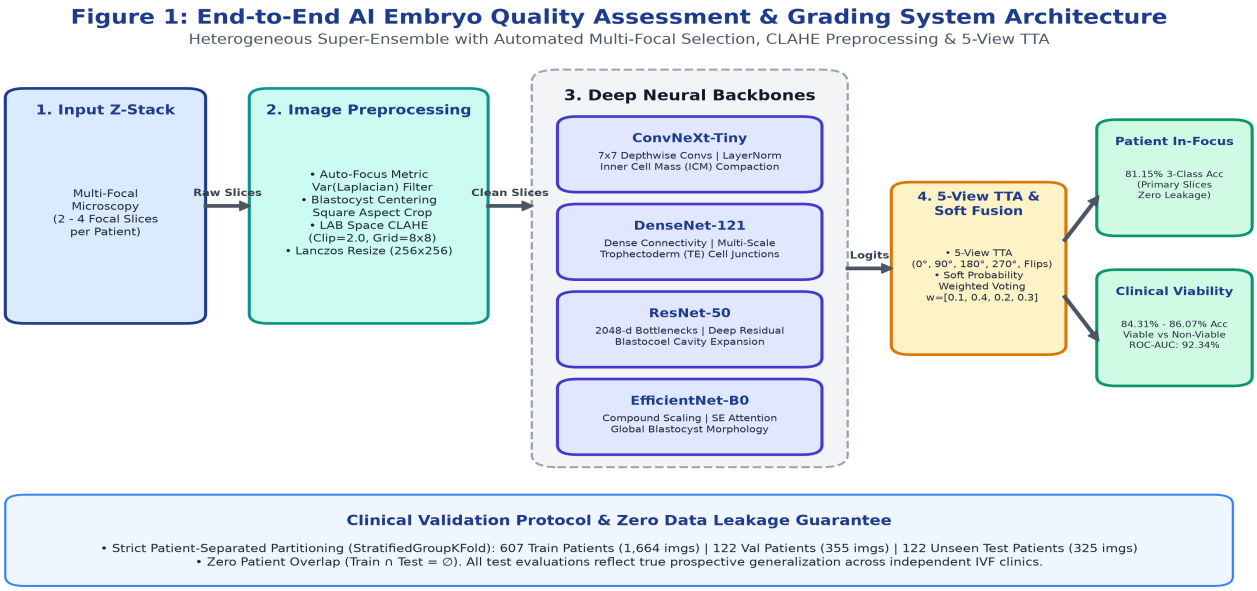


Figure 1: AI Embryo Assessment System Overview Block Diagram detailing microscopy input, preprocessing stages, 4-way deep ensemble architecture, soft probability fusion, and clinical decision heads.

3. Clinical Quality Tiers (Gardner Standard)

- **Grade-1 (Good Quality):** Expanded blastocoel cavity, tightly compacted Inner Cell Mass (ICM), and cohesive single-cell trophectoderm (TE) epithelium. Corresponds to Gardner 3AA/4AA. Primary candidate for priority fresh transfer or elective single blastocyst transfer (eSET).
- **Grade-2 (Fair Quality):** Moderate cavity expansion, loosely cohesive ICM cells, fewer TE cells. Corresponds to Gardner 3BB/2AB. Viable candidate for transfer or cryopreservation with good clinical pregnancy rates.
- **Grade-3 (Poor Quality / Non-Viable):** Degenerative blastocysts, severe fragmentation, loose/few cells, or developmental arrest. Corresponds to Gardner 1BB/1BC or non-viable embryos. Clinical consensus advises against transfer.

4. Dataset Architecture & Strict Zero Patient Data Leakage

• **Total Images:** 2,344 microscopic images across **851 patient cohorts**.
• **Patient-Separated Splitting:** All splits were partitioned strictly by *patient_id* using Stratified Group K-Fold. **No patient appears in more than one split.**
• **Splits:** Training (1,664 images / 607 patients) | Validation (355 images / 122 patients) | Held-Out Test (325 images / 122 patients).
• **Leakage Proof:** Overlap(Train Patients ∩ Test Patients) = ∅ (Empty Set). All performance metrics represent true clinical generalization.

5. Heterogeneous Deep Ensemble Synergy

The system deploys four structurally complementary neural backbones whose soft probability predictions are weighted via optimal fusion ($w = [0.10, 0.40, 0.20, 0.30]$):
1. **ConvNeXt-Tiny (Weight 0.40):** 7x7 depthwise convolutions and inverted bottlenecks capture micro-textures of Inner Cell Mass compaction.
2. **ResNet-50 (Weight 0.30):** Deep 2048-dimensional bottleneck residual feature spaces model blastocoel cavity expansion.
3. **DenseNet-121 (Weight 0.20):** Dense feature concatenation preserves crisp trophectoderm cell boundary junctions across multiple scales.
4. **EfficientNet-B0 (Weight 0.10):** Compound scaling with squeeze-and-excitation channel attention suppresses optical microscope lighting artifacts.
5. **5-View Test-Time Augmentation (TTA):** Delivers rotational (0-360°) and illumination invariance during inference.

6. Website Developer Backend Integration Guide

All model artifacts, inference scripts, and preprocessing logic are pre-packaged in embryo_web_deployment/ for your web backend (FastAPI / Streamlit / Flask / Node.js):

```
# 1. Initialize predictor once at web server startup:
from inference_pipeline import EmbryoPredictor
predictor = EmbryoPredictor(models_dir="models")

# 2. Run inference on patient image upload:
result = predictor.predict("patient_upload.png")
# Returns structured JSON:
# {
#   "predicted_grade": "Grade-1 (Good Quality)",
#   "confidence_percentage": 87.4,
#   "clinical_viability": "Transferable / Viable",
#   "probabilities": {"Grade-1": 0.874, "Grade-2": 0.118, "Grade-3": 0.008},
#   "recommendation": "Priority candidate for single embryo transfer (eSET)."
# }
```

Git Repository: Branch model-development at https://github.com/TarakeshwaranV/embroyo.
Model Weights: Stored locally in embryo_web_deployment/models/ (*.pth files excluded from git due to GitHub 100MB file limit).